MS1 Review: P14 (AuroraCast)

1. Summary

AuroraCast forecasts the planetary Kp-index for the next 3-hour interval, starting at 00. It uses only data published before the issued timestamp t, plus the probability of a G1+ Storm (Kp>5). 1-minute solar wind and magnetic field data from the L1 spacecraft (NOAA SWPC) are used as inputs, with historical backfill from NASAs OMNIWeb (around 82k labelled intervals). The doc intends to compare a storm classifier and a gradient-boosted regressor against persistence and only registering if they win. Success is tiered, with the minimum defined as beating persistence on MAE and on storm recall at equal precision. Target MAE is 10% below persistence and storm recall >0.5 at prec > 0.4.

2. Strengths

Leakage handling is very specific and detailed. The proposal is aware of risks such as data delay and L1-to-Earth shift. Labels cannot be derived from (satellite) inputs, as they're collected from ground magnetometers. The baseline is defined precisely, and NOAA's own forecast is logged as an external reference. The rare class is handled with class weights and evaluated with PR-AUC and per-storm-event hit rate instead of accuracy. The doc also clearly states that the live feed only keeps around 24h, so the poller starts the history from the first run.

3. Concerns

Train/Serve skew: Historical data is processed OMNI data with definitive Kp values, while the live inference uses raw realtime SWPC data and nowcast Kp. The doc mentions this as a "small, known difference" but defines no clear measurement strategies, which could lead to MAE overstating live performance.

Eval Size:
Storms are a few % of intervals and come in clusters. The 2024-2025 test set holds few independent events and recall at prec > 0.4 will be noisy. The doc also describes its targets as estimated to be "re-justified" after EDA; could lead to "moving goalposts".

Originality: While it was checked against the named sources, NOAA already publishes its own forecast.

4. Suggestion

Add NOAA's Kp forecast part of the success criterion. Report storm recall at equal precision against NOAA on the dashboard for a clear comparison! Also, freeze thresholds before touching the test set and state the number of (independent) storms in it.